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SIR – Julia, .

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3

SIR

SIR (Kermack & McKendrick, 1927)
(,); **I (Infectious)** —
(Recovered) — (

: **S (Susceptible)** — (); **R** ().

(,): $N = S + I + R = \text{const.}$:

. : () () .

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$$\begin{cases} \frac{dS}{dt} = -\beta \cdot c \cdot \frac{I}{N} \cdot S \\ \frac{dI}{dt} = \beta \cdot c \cdot \frac{I}{N} \cdot S - \gamma \cdot I \\ \frac{dR}{dt} = \gamma \cdot I \end{cases}$$

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; (R_0 < 1):

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$$R_0 = \frac{c \cdot \beta}{\gamma}$$

. ($R_0 > 1$):

$$HIT = 1 - \frac{1}{R_0}$$

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$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = \delta xy - \gamma y \end{cases}$$

: * **x** —

(,); * **y** —

(,); * (alpha)

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$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = \delta xy - \gamma y \end{cases}$$

: * **x** —

(,); * **y** —

(,); * (alpha)

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(); * (beta) —

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* (delta) —

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$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = \delta xy - \gamma y \end{cases}$$

: * **x** —

(,); * **y** —

(,); * (alpha)

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(); * (beta) —

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* (delta) —

(); * (gamma) —

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sir_ode.jl, SIR. 6 ,
(. 1).

```
Для проверки: using DrWatson, DifferentialEquations, Plots  
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project$ nano scripts/si  
r_ode.jl  
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project$ julia --project  
=. scripts/sir_ode.jl
```

1: sir_ode.jl

lv_ode.jl, — 6 ,
(. 2).

```
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project$ julia --project=. script s/lv_ode.jl
```

2: lv_ode.jl

sir_ode.jl lv_ode.jl

(. 3).

```
avbogdanyuk@COMPUTER: ~ X + v
GNU nano 7.2 scripts/sir_ode.jl *
## Модель SIR
#Модель SIR делит всю популяцию на три взаимосвязанные группы (компарменты), что отражено в её названии:
#- S - Susceptible (Восприимчивые): люди, которые не болели, не имеют иммунитета и могут заразиться.
#- I - Infectious (Инфицированные/Заразные): люди, которые в данный момент больны и могут передавать инфекцию.
#- R - Recovered (Выздоровевшие/Удаленные): люди, которые переболели и приобрели иммунитет (или умерли). Они больше не участв
using DrWatson
@quickactivate "project"

using DifferentialEquations
using SimpleDiffEq
using Tables
using DataFrames
using StatsPlots
using LaTeXStrings # Для красивого отображения формул на графиках
using Plots
using BenchmarkTools

script_name = splitext(basename(PROGRAM_FILE))[1]
mkpath(plotsdir(script_name))
mkpath(datadir(script_name))

function sir_ode!(du, u, p, t)
    (S, I, R) = u
    (β, c, γ) = p
```

3:

tangle.jl

sir_ode.jl lv_ode.jl

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(. 4).

```
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project$ nano scripts/tangle.jl
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project$ julia --project=. script
s/tangle.jl scripts/sir_ode.jl
Precompiling Literate finished.
 2 dependencies successfully precompiled in 7 seconds. 17 already precompiled.
Генерация из: scripts/sir_ode.jl
[ Info: generating plain script file from '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl'
✓ Чистый скрипт: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/scripts/sir_ode/sir_ode.jl
[ Info: generating markdown page from '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/markdown/sir_ode/sir_ode.qmd'
✓ Quarto: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/markdown/sir_ode/sir_ode.qmd
[ Info: generating notebook from '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/scripts/sir_ode.jl'
[ Info: writing result to '~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/notebooks/sir_ode/sir_ode.ipynb'
✓ Notebook: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab02/project/notebooks/sir_ode/sir_ode.ipynb

Готово! Все файлы созданы.
```

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(Jupyter Notebook, SIR, —, Julia, Quarto)